



Material Declaration Sheet

Vishay Inter-technology

Date 12/8/2023

Part / Product Family Details

Part / Family Series	RoHS Compliance Status	RoHS Compliance Date Code dd-mm-yyyy	Total product Weight (mg)	Value Range	3rd Party Lab ICP Test Report Available	Manufacturing Location	Number of Exemptions Used
CRCW0603 e3 (10R to 1M)	YES WITH EXEMPTION	16-Oct-2004	2.04	10R to 1M	Yes	Israel	1

Technical Information

Terminal Plating / Grid Array Material	Terminal Base Alloy	JESD-97 Pb-Free Material Code Marking	J-STD-20D MSL Rating	Reflow Peak Process Body Temperature	Reflow Maximum number of cycles	Reflow Max.Time at Peak Temperature (sec)	Soldering Compatibility (SnPb/Pb-Free)
Tin (Sn)	Other	e3	1	260 ° C	3	40	Backward & Forward

Material Composition

Homogenous Material Name	Material Classification	Substance Name	CAS number	Weight of Substance (mg)	With respect to Homogenous Material		% with respect to Total Product Weight	RoHS Exemptions Used
					%	ppm		
Bottom Conductor	Ceramics / glass	Glass, oxide	65997-17-3	0.00582	3.44	34364	0.01	0 - None
		Silver	7440-22-4	0.00562	96.56	965636	0.28	0 - None
Ceramic Substrate	Ceramics / glass			1.81154				
		Aluminium oxide	1344-28-1	1.73908	96.00	960001	85.25	0 - None
		Calcium oxide	1305-78-8	0.00362	0.20	1998	0.18	0 - None
		Magnesium oxide	1309-48-4	0.02355	1.30	13000	1.15	0 - None
		Silicon dioxide	14808-60-7	0.04529	2.50	25001	2.22	0 - None
Front side	Other special metal			0.00058				
		Chromium	7440-47-3	0.00022	37.93	379310	0.01	0 - None
		Copper	7440-50-8	0.00009	15.52	155172	0.00	0 - None
		Nickel	7440-02-0	0.00027	46.55	465517	0.01	0 - None
Nickel plating	Nickel alloys			0.07417				
		Miscellaneous	System	0.00007	0.10	1000	0.00	0 - None
Overcoat	Lacquers	Nickel	7440-02-0	0.07410	99.90	999000	3.63	0 - None
				0.02772				
		Carbon black	1333-86-4	0.00180	6.49	64935	0.09	0 - None
Passivation	Ceramics / glass	Epoxy resin	25068-38-6	0.01400	50.51	505051	0.69	0 - None
		Silicon dioxide	60676-86-0	0.01192	43.00	430014	0.58	0 - None
				0.00750				
Resistive element	Ceramics / glass	Lead silicate [Glass]	65997-18-4	0.00749	99.90	999000	0.37	7(c)-I
		miscellaneous	System	0.00001	0.10	1000	0.00	0 - None
Tin plating	Other special metal			0.01371				
		Lead silicate [Glass]	65997-18-4	0.00480	35.01	350109	0.24	7(c)-I
		Palladium	7440-05-3	0.00069	5.03	50328	0.03	0 - None
		Ruthenium oxide	12036-10-1	0.00363	26.48	264770	0.18	0 - None
		Silver	7440-22-4	0.00459	33.48	334792	0.23	0 - None
Top Conductor	Ceramics / glass			0.07417				
		miscellaneous	System	0.00007	0.10	1000	0.00	0 - None
		Tin	7440-31-5	0.07410	99.90	999000	3.63	0 - None
				0.02479				
		Glass, oxide	65997-17-3	0.00037	1.49	14925	0.02	0 - None
		Palladium	7440-05-3	0.00012	0.48	4841	0.01	0 - None
		Silver	7440-22-4	0.02430	98.02	980234	1.19	0 - None

EU-RoHS Directive-2011/65/EU MCV of 0.1% by mass (1000 PPM) in homogeneous material for: Lead (Pb), Mercury, Hexavalent Chromium, Polybrominated Biphenyls (PBB), Polybrominated Diphenyl Ethers (PBDE) and MCV of 0.01% by mass cadmium

Exemption Used 7(c)-I - Electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectronic devices, or in a glass or ceramic matrix compound

Note :- (i) All information is based on data received from our vendors & subjected to change without prior notice.
(ii) Substance weight are derived from MSDS.

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